

Chapter 3

Real functions and Continuity

① Boundedness of the function:

1) f is bounded below:

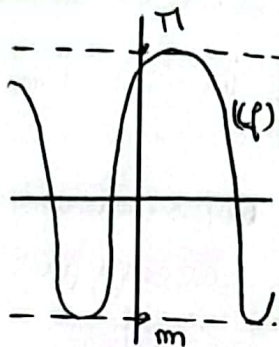
$$\forall x \in \mathbb{R}, \exists m \in \mathbb{R} \text{ s.t. } f(x) \geq m$$

2) f is bounded above:

$$\forall x \in \mathbb{R}, \exists M \in \mathbb{R} \text{ s.t. } f(x) \leq M$$

3) f is bounded:

$$\forall x \in \mathbb{R}, \exists M > 0 \text{ s.t. } |f(x)| \leq M$$



② Monotone functions:

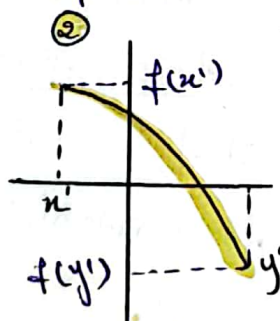
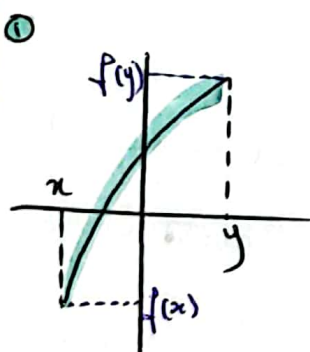
Let $f: D_f \rightarrow \mathbb{R}$. We say that:

1) f is increasing: $\forall x, y \in D_f: x \leq y \Rightarrow f(x) \leq f(y)$

2) f is decreasing: $\forall x, y \in D_f: x \leq y \Rightarrow f(x) \geq f(y)$

3) f is monotone if: $\rightarrow f$ is increasing

$\rightarrow f$ is decreasing

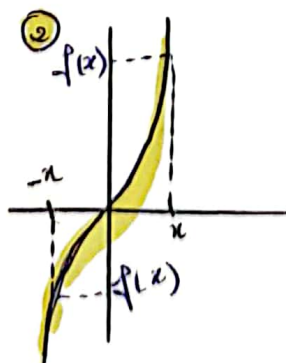
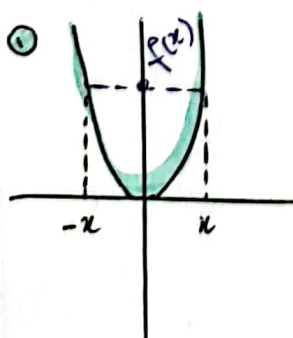


3) parity and periodicity:

Let $f: I \rightarrow \mathbb{R}$, we say that:

1) f is even: $\forall x \in I, f(-x) = f(x)$

2) f is odd: $\forall x \in I, f(-x) = -f(x)$



periodic function:

$f: \mathbb{R} \rightarrow \mathbb{R}$ is a function and $T > 0$

f is periodic with period $T \Rightarrow \forall n \in \mathbb{R}$ we have:

$$f(x+T) = f(x)$$

example:

$$f(x) = \cos x, T = 2\pi$$

$$f(x+2\pi) = \cos(x+2\pi) = \cos x = f(x)$$

$$f(x) = \sin x, T = 2\pi$$

$$f(x+2\pi) = \sin(x+2\pi) = \sin x = f(x)$$

4) Limits:

Definitions: limit at a point

$f: I \rightarrow \mathbb{R}$ be a function, let $x_0 \in \mathbb{R}$ and $\ell \in \mathbb{R}$

f has a limit ℓ in x_0 if: $\forall \epsilon > 0, \exists$

$$\delta_\epsilon > 0, \forall x \in I$$

$$(|x - x_0| < \delta_\epsilon) \Rightarrow (|f(x) - \ell| < \epsilon)$$

we write:

$$\lim_{x \rightarrow x_0} f(x) = \ell$$

example:

prove that $\lim_{n \rightarrow 0} \sqrt{n} = 0$: $n \gg 0$

let $\epsilon > 0$ s.t. $|f(x) - \ell| < \epsilon$

$$\Rightarrow |\sqrt{x} - 0| < \epsilon \Rightarrow \sqrt{x} < \epsilon \Rightarrow x < \epsilon^2$$

$$|x - 0| < \delta \Rightarrow |x| < \delta \Rightarrow x < \delta$$

Then: $\exists \delta_\epsilon = \epsilon^2 > 0$

Other definitions:

① $\lim_{x \rightarrow x_0} f(x) = +\infty$

$\Leftrightarrow \forall A > 0; \exists \delta > 0; |x - x_0| < \delta \Rightarrow f(x) > A$

② $\lim_{x \rightarrow x_0} f(x) = -\infty$

$\Leftrightarrow \forall A > 0; \exists \delta > 0; |x - x_0| < \delta \Rightarrow f(x) < -A$

③ If $x_0 = \pm \infty$

- $\lim_{x \rightarrow +\infty} f(x) = +\infty$
 $\Leftrightarrow \forall A > 0, \exists B > 0; x > B \Rightarrow f(x) > A$
- $\lim_{x \rightarrow -\infty} f(x) = +\infty$
 $\Leftrightarrow \forall A > 0, \exists B > 0, x < -B \Rightarrow f(x) > A$

5) Properties:

P₁: f has a limit $l \Rightarrow l$ is unique

P₂: $\lim_{x \rightarrow x_0} f(x) = l_1; \lim_{x \rightarrow x_0} g(x) = l_2$

$\rightarrow \lim_{x \rightarrow x_0} (\lambda f)(x) = \lambda l_1$

$\rightarrow \lim_{x \rightarrow x_0} (f+g)(x) = l_1 + l_2$

$\rightarrow \lim_{x \rightarrow x_0} (f \cdot g)(x) = l_1 \cdot l_2$

$\rightarrow \lim_{x \rightarrow x_0} \frac{1}{f(x)} = \frac{1}{l_1}; l_1 \neq 0$

P₃: $\rightarrow$ ① $f(x) \leq g(x)$ and $\lim_{x \rightarrow x_0} f(x) = +\infty$

Then: $\lim_{x \rightarrow x_0} g(x) = +\infty$

② $f(x) \leq g(x) \leq h(x)$
 and $\lim_{x \rightarrow x_0} f(x) = +\infty$ and $\lim_{x \rightarrow x_0} h(x) = +\infty$
 Then: $\lim_{x \rightarrow x_0} g(x) = +\infty$

⑥ Continuity:

① f is continuous at $x_0 \in I, \text{ if } I = I$:

$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} f(x) = f(x_0)$

② f is continuous over I : f is continuous at every point of I .

Propositions:

$f: I \rightarrow \mathbb{R}; g: I \rightarrow \mathbb{R}$

f and g are continuous at $x_0 \in I$ Then:

① $f+g$ is continuous function at x_0

② λf is continuous function at $x_0, \forall \lambda \in \mathbb{R}$

③ $f \cdot g$ is continuous function at x_0

④ $f(x) \neq 0$ Then $\frac{1}{f}$ is continuous function at x_0

⑦ Continuous Extension:

Let $f: I - \{x_0\} \rightarrow \mathbb{R}$ be a fund.

$\hookrightarrow f$ is extendable by continuity at x_0

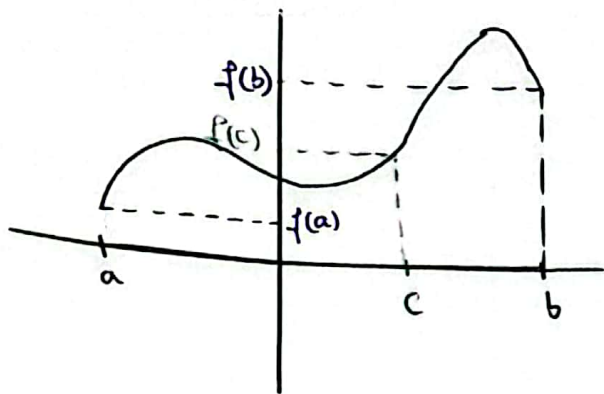
if $\lim_{x \rightarrow x_0} f(x) = l < \infty$

$\hookrightarrow$ we say f admits an extension by continuity at $x_0 = 0$.

⑧ Intermediate value Theorem:

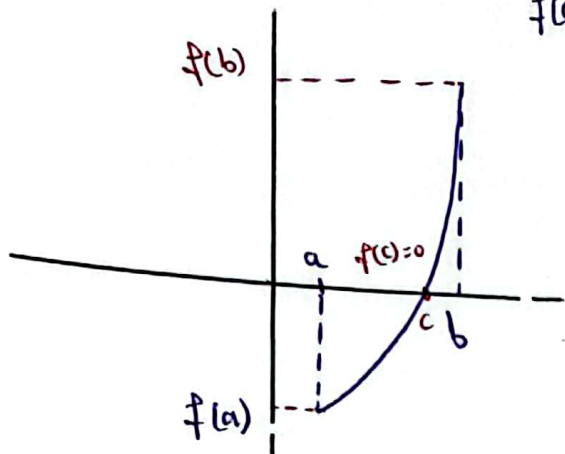
$f: [a; b] \rightarrow \mathbb{R}$ (f continuous).

$\left(\forall y \in \mathbb{R} \right. \left. \begin{matrix} f(a) \leq y \leq f(b) \end{matrix} \right) \Rightarrow \left(\exists c \in [a; b] \right. \left. \begin{matrix} \text{st: } y = f(c) \end{matrix} \right)$



Corollary:

If $(f(a) \times f(b) < 0) \Rightarrow (\exists c \in]a, b[\text{ st } f(c) = 0)$



Theorem:

If f is continuous in $[a, b]$, then f is bounded $[a, b]$ (has a $\begin{cases} \rightarrow \text{max} \\ \rightarrow \text{min} \end{cases}$).

$\exists \pi > 0 \text{ st } |f(x)| \leq \pi$.

⑨ Monotone and bijective function:

$f: I \rightarrow \mathbb{R}$ be a $\begin{cases} \rightarrow \text{continuous} \\ \rightarrow \text{strictly monotone} \end{cases}$

① $f: I \rightarrow J: f(I)$ establishes a bijection

2) The Inverse function $f^{-1}: J \rightarrow I$
 f^{-1} is continuous and strictly monotone in J

So: $\begin{cases} \text{If } f \text{ is } \nearrow \sim f^{-1} \text{ is } \nearrow \\ \text{If } f \text{ is } \searrow \sim f^{-1} \text{ is } \searrow \end{cases}$

